

Approximate integrability, KAM, and the value of explicit constants of motion

Working notes

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Abstract

These notes summarise the distinction between *formal* (perturbative) and *symmetry-based* constructions of approximate constants of motion in Hamiltonian systems, locate where each is valid, and collect the arguments for why the construction (or proven non-existence) of explicit, global constants of motion is not made redundant by KAM theory or by near-identity transforms. The intended use is as a reference and a source of deployable counter-arguments in seminars and refereeing.

1 The notion: integrability up to $O(\varepsilon^N)$

Take an integrable Hamiltonian H_0 on a $2n$ -dimensional symplectic phase space, with n functionally independent integrals $C_1, \dots, C_n$ in involution ($\{C_i, C_j\} = 0$). Add a perturbation,

$$H = H_0 + \varepsilon H_1, \quad 0 < \varepsilon \ll 1. \quad (1)$$

One may try to deform the integrals as $C_i^{\text{new}} = C_i + \varepsilon C_i' + \varepsilon^2 C_i'' + \dots$ and ask whether the corrections can be chosen so that

$$\{C_i^{\text{new}}, C_j^{\text{new}}\} = O(\varepsilon^{N+1}), \quad \{C_i^{\text{new}}, H\} = O(\varepsilon^{N+1}), \quad (2)$$

for some $N \geq 1$. When such a deformation succeeds, one calls the system *integrable up to order ε^N* or says one has constructed *formal (approximate) integrals of motion*. Classical entry points:

- **Constructive (perturbation theory):** G.D. Birkhoff, *Dynamical Systems* (AMS, 1927); F.G. Gustavson, *Astron. J.* **71**, 670 (1966).
- **Obstruction (non-existence):** H. Poincaré, *Les méthodes nouvelles de la mécanique céleste*, Vol. I (1892), Ch. V.
- **Divergence of the formal series:** C.L. Siegel, *Ann. Math.* **42**, 806 (1941).
- **What survives the divergence:** N.N. Nekhoroshev, *Russ. Math. Surveys* **32**(6), 1 (1977).
- **Modern survey:** Arnold, Kozlov & Neishtadt, *Mathematical Aspects of Classical and Celestial Mechanics* (Springer).

2 Two distinct constructions

(A) Perturbative normal form (Birkhoff–Gustavson)

One obtains C'_i by inverting $\{\cdot, H_0\}$. In action–angle coordinates of H_0 this is, on Fourier modes, multiplication by $i\omega(I)\cdot k$, and the inverse carries divisors

$$\frac{1}{\omega(I)\cdot k}. \quad (3)$$

Two distinct failures follow:

- *Finite-order singularity.* Already at first order, C'_i is singular on the resonant set $\{\omega(I)\cdot k = 0\}$. The construction is valid only off resonances; on resonances the correction blows up.
- *Generic divergence.* Even off resonances, the full series in ε generically diverges (Siegel). The formal integrals are asymptotic, not convergent.

The deep statement that no *global, single-valued* formal integral beyond H exists, under non-degeneracy and genericity of H_1 , is Poincaré’s theorem on the non-existence of additional analytic integrals.

Special case: non-resonant elliptic equilibrium. If H_0 is the quadratic part near an elliptic fixed point with frequencies ω that are rationally independent constants, the divisors $\omega\cdot k$ never vanish: the normalization proceeds to *all* orders as formal power series, and one obtains n formal integrals in involution. This makes *every* such system formally integrable locally — which is precisely why “formal integrability” is, by itself, weak: the series diverge, the integrals do not exist as honest functions, and chaos coexists with them. Nekhoroshev quantifies the practical lifetime: exponentially long-time conservation, not exact conservation.

(B) Symmetry-based construction (Killing–Yano)

In the geometric setting relevant to general relativity, the correction C'_i is not obtained by inverting anything. One *writes it down* explicitly from the spacetime’s hidden symmetries — typically a Killing(–Yano) tensor of the background — and verifies

$$\{C_i + \varepsilon C'_i, H\} = O(\varepsilon^2) \quad (4)$$

as a covariant tensorial identity, valid pointwise on phase space. The reason this works globally is that the resonant part of the source $\{C_i, H_1\}$ — the very part that would generate a small divisor in (A) — *vanishes identically* by Killing-tensor identities. The homological equation then admits a smooth, bounded solution *everywhere*, resonances included, and the near-involution $\{C_i^{\text{new}}, C_j^{\text{new}}\} = O(\varepsilon^2)$ is an algebraic statement on phase space, orbit-independent, hence true on resonant orbits too.

The obstruction to going from $O(\varepsilon^2)$ to $O(\varepsilon^3)$ is then *geometric* (does the background admit the requisite higher-rank Killing(–Yano) structure?), not a small-divisor obstruction. This is the sense in which (B) is structurally cleaner: global and exact-to-order, underwritten by spacetime geometry. The price is that (B) is not generic; it exists only when the unperturbed background carries the symmetry.

Local vs. global, summarised

	(A) Normal form	(B) Symmetry-based
Mechanism	Invert $\{\cdot, H_0\}$ via Fourier; small divisors.	Explicit Killing-tensor construction; no inversion.
Validity in phase space	Off resonances only; singular on $\omega \cdot k = 0$.	Pointwise everywhere; resonances included.
Convergence in ε	Generically divergent series (Siegel).	Exact-to-order; controlled error at all ε in regime.
Generic obstruction	Resonance web; Poincaré theorem.	Existence of the right Killing(–Yano) tensor.
Coordinate dependence	Coordinate-dependent formal series.	Covariant tensorial identity.

3 What KAM actually says — and what it does not

The Kolmogorov–Arnold–Moser theorem, under (Kolmogorov) non-degeneracy $\det \partial^2 H_0 / \partial I^2 \neq 0$, smoothness, and small enough $\varepsilon < \varepsilon_*$, asserts: the strongly Diophantine invariant tori of H_0 *persist* as slightly deformed invariant tori of $H_0 + \varepsilon H_1$, forming a Cantor set of positive measure in action space. It does *not* assert that the perturbed system is integrable. Generically the perturbed system is *non-integrable* (Poincaré), and KAM is consistent with this: the chaotic regions live in the open dense complement of the surviving tori. “KAM tells us we stay integrable” is a category error.

What KAM does *not* give:

- No explicit constants of motion. No actions, no frequencies, no ISCO correction, no separatrix shift, no input for a waveform pipeline.
- No confinement of orbits for $n \geq 3$ (Arnold diffusion; V.I. Arnold, *Sov. Math. Dokl.* **5**, 581 (1964)). KAM tori are n -dimensional in a $(2n - 1)$ -dimensional energy surface and do not topologically separate it for $n \geq 3$. *Caveat for EMRIs*: the Killing vectors of Kerr ($\partial_t, \partial_\phi$) and, where applicable, $\|S\|$ reduce the effective phase space to $n = 2$, so KAM curves do confine motion. Chaos is then *bounded* but not absent: surviving KAM curves coexist with resonance islands and stochastic layers (Hénon–Heiles picture). Positive Lyapunov exponents and chaotic Poincaré sections in this regime remain fully meaningful and signal genuine non-integrability of the reduced system.
- Nothing about the resonant set, where the orbits of physical interest (resonance crossings in inspirals) actually live.
- Nothing about non-integrability, chaos, or Lyapunov behaviour — it is silent on these.
- No applicability to dissipative dynamics. KAM is a conservative statement; self-force is not.
- No usable threshold. ε_* depends on Diophantine constants and smallest divisors; in concrete celestial-mechanics problems it is typically many orders of magnitude below physical ε (Celletti–Chierchia).

Where KAM *is* genuinely strong (concede explicitly)

In *conservative*, $n = 2$, *provably-small- ε* systems for *qualitative* questions about long-time confinement near the unperturbed structure. In that narrow setting, an additional construction of

approximate integrals is largely redundant for qualitative purposes. None of these conditions are satisfied in the EMRI/IMRI/tidal context.

4 Defending explicit constants against common critiques

The single most useful move is to refuse to defend on the critic’s terms and to reframe: KAM and near-identity transforms answer *existence* questions (do tori persist as a set?); explicit-constant work answers *construction* questions (what are the constants, and where don’t they exist?). These are different questions, and the first does not substitute for the second.

“KAM says you stay integrable anyway.” Category error. KAM asserts persistence of a Cantor set of tori in a *non-integrable* system. Chaos coexists with the surviving tori; KAM gives no constants, no frequencies, no information at resonances. In the EMRI setting, where Kerr’s Killing vectors reduce the effective phase space to $n = 2$ and KAM curves *do* confine, this only forces chaos to be *bounded*—resonance islands and stochastic layers persist between surviving KAM curves (Hénon–Heiles picture), and explicit positive Lyapunov / Poincaré-section evidence (as in tidal Kerr) directly demonstrates non-integrability of the reduced system. For higher- n problems with no isometric reduction (the Solar System, multi-body configurations), Arnold diffusion further weakens any naive “trapped near tori” intuition.

“Birkhoff–Gustavson / near-identity transforms do this formally to all orders.” Formal series with small divisors, singular on the resonant set at finite order, generically divergent at infinite order. The symmetry-based construction is exact-to-order, global, with no divisors. Different machines, different domains of validity.

“For small enough ε KAM rescues you.” The threshold ε_* is generically unspecified, and in concrete systems extremely small. More importantly, the empirical record refutes this reflex: the Solar System has $\varepsilon \sim 10^{-3}$ (Jupiter) and is demonstrably chaotic (Laskar, *Nature* **338**, 237 (1989); Laskar–Gastineau, *Nature* **459**, 817 (2009): Mercury orbital instability at the Gyr level). If the canonical small- ε Hamiltonian system in Nature is chaotic, “small $\varepsilon \Rightarrow$ KAM saves us” is empirically bankrupt. What actually limits KAM in practice is overlapping resonances—Chirikov’s criterion (*Phys. Rep.* **52**, 263 (1979)).

“ ε is small in your problem anyway.” Not in the systems of astrophysical interest:

- IMRIs: mass ratio $10^{-2} - 10^{-4}$, two to four orders above EMRI.
- Main-sequence stars or white dwarfs around an SMBH: tidal deformability $\sim (R/M)^5$, many orders of magnitude larger than for a stellar BH. The quadrupolar / spin-squared couplings are not “small” in any meaningful sense.

Small- ε KAM is not applicable to these problems even on its own terms.

“It’s only leading order; small effect.” Linear-in-spin is the dominant spin contribution to inspiral phasing, i.e. the regime that LISA must model. And the obstruction at quadratic order is itself a structural finding: it identifies the geometric (Killing–Yano) condition under which spin terms remain integrable, and otherwise generate chaos.

“Self-force is dissipative, so neither KAM nor your conservative result apply.” KAM does not apply, correct. The conservative construction is the *backbone* around which two-timescale frameworks (Hinderer & Flanagan, *Phys. Rev. D* **78**, 064028 (2008); Flanagan & Hinderer, *Phys. Rev. Lett.* **109**, 071102 (2012)) are built. The explicit constants are exactly the conservative input

that the dissipative expansion needs. The non-existence/chaos results then say precisely where that backbone breaks (transient resonances), which is itself necessary input.

No constants $\Rightarrow$ no quadrature. The Liouville–Arnold equivalence is hard: n commuting integrals on a $2n$ -dim symplectic manifold $\Leftrightarrow$ solvable by quadratures, with action–angle coordinates on bounded tori. Without the integrals one loses closed-form orbit solutions, decoupled degrees of freedom, well-defined frequencies, action–angle coordinates, and the guiding-centre backbone of any two-timescale expansion. KAM persistence of *abstract* tori as a Cantor set does not supply the explicit map (action, angle) $\leftrightarrow$ phase-space point that a waveform template bank requires.

Local NIT validity is moot once chaos is exhibited. If chaotic orbits are demonstrated (positive Lyapunov, dense Poincaré sections), the system is non-integrable, full stop. NIT being applicable in non-resonant patches is consistent with this; it does not save global integrability. An opponent invoking KAM is implicitly conceding the chaos and asking that one ignore the orbits that visit it.

Closing positioning

KAM and near-identity transforms describe *what survives*; explicit-constant work describes *what is*. The two sit at different levels: persistence theorems sit *downstream* of the explicit construction (which provides actions and frequencies as honest functions) and *upstream* of nothing one actually computes with. In the EMRI program in particular, the explicit constants and their non-existence theorems are not redundant with KAM; they supply content KAM cannot produce.

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